

# AWS Production Setup & Deployment Guide

IAM + S3 + CloudFront + EC2 + GoDaddy + Docker/Nginx

English Edition

```
GoDaddy DNS
|
+--> bookkeepersmarketplace.store -----> Frontend :3000
+--> admin.bookkeepersmarketplace.store -> Admin :3001
+--> api.bookkeepersmarketplace.store ---> Backend :5000
      |
      Elastic IP
      |
      AWS EC2 + Nginx
      |
      MongoDB / Stripe / S3 + CloudFront
```

Reusable reference based on the Bookkeepers Marketplace deployment

## Quick Reference - Current Project

**Security:** This document intentionally contains no real passwords, secret access keys, or Stripe secrets. Keep secrets in environment variables / a secrets manager and never commit them to Git.

| Item        | Current value                                                      |
|-------------|--------------------------------------------------------------------|
| Root domain | bookkeepersmarketplace.store                                       |
| Website     | www.bookkeepersmarketplace.store +<br>bookkeepersmarketplace.store |
| Admin       | admin.bookkeepersmarketplace.store                                 |
| Backend     | api.bookkeepersmarketplace.store                                   |
| AWS Region  | us-east-2 (Ohio)                                                   |
| EC2 name    | bookkeepers-marketplace-production                                 |
| EC2 type    | t3.small - 2 vCPU, 2 GB RAM                                        |
| OS          | Ubuntu Server 26.04 LTS, x86_64                                    |
| EBS         | 30 GB gp3                                                          |
| Elastic IP  | 3.151.181.212                                                      |
| S3 bucket   | bookkeepers-marketplace-prod-storage                               |
| CloudFront  | d26wfj7v7cajxb.cloudfront.net                                      |

### Full Flow in One Line

AWS Account -> Root MFA -> IAM User -> S3 -> CloudFront OAC -> EC2 -> Elastic IP -> GoDaddy DNS -> SSH -> Docker/Nginx -> Deploy -> SSL -> Stripe Webhook

## 1. AWS Account, Billing, and Root Security

1. Create the AWS account with the client/company email. The client should remain the account owner.
2. The client completes billing/payment setup. A developer should not collect card, CVV, or bank credentials.
3. Enable MFA on the AWS root account.
4. Do not use the root account for everyday deployment. Use a separate IAM user or IAM Identity Center access.

**Rule:** Never request or share the root password, root MFA code, root access keys, or billing credentials.

## 2. Create a Developer IAM User

5. AWS Console -> IAM -> Users -> Create user.
6. Username: saurav-developer (or a project-specific developer name).
7. Enable AWS Management Console access.
8. Use an autogenerated temporary password and require a password change on first sign-in.
9. AdministratorAccess can be used temporarily for initial infrastructure setup. Reduce permissions to least privilege after setup.
10. After creation, securely provide the IAM sign-in URL, username, and temporary password to the developer.

### IAM sign-in details

IAM sign-in URL: `https://ACCOUNT-ID.signin.aws.amazon.com/console`  
 Username: saurav-developer  
 Temporary password: \*\*\*\*\*

**Recommended:** IAM Identity Center is the preferred long-term option for workforce users. A separate IAM user is still a simple option for a small/client-managed setup.

## 3. S3 - Private Image, Video, and Document Storage

11. AWS Console -> S3 -> Create bucket.
12. Bucket type: General purpose.
13. Use a globally unique bucket name. Current example: bookkeepers-marketplace-prod-storage.
14. Keep the bucket in the same region as EC2 when practical. Current project: us-east-2 (Ohio).
15. Object Ownership: ACLs disabled / Bucket owner enforced.
16. Keep Block Public Access fully enabled.
17. Keep default encryption enabled (SSE-S3 is fine to start).
18. Use object prefixes such as uploads/, documents/, and videos/ as needed.

**Expected behavior:** A direct S3 URL returning AccessDenied is normal for a private bucket. Use CloudFront for public delivery.

## 4. How to Create S3 Access Keys

19. IAM -> Users -> developer/s3-uploader user -> Security credentials.
20. Access keys -> Create access key.
21. Choose Application running outside AWS or the appropriate use case.
22. Create the key to receive an Access key ID and Secret access key.
23. The secret access key is shown in full only when it is created. Save it securely at that time.

```

AWS_ACCESS_KEY_ID=AKIA...
AWS_SECRET_ACCESS_KEY=...
AWS_BUCKET_NAME=bookkeepers-marketplace-prod-storage
AWS_REGION=us-east-2
AWS_PUBLIC_BASE_URL=https://d26wfj7v7cajxb.cloudfront.net

```

**Production best practice:** If the backend runs on EC2, prefer an EC2 IAM Role for S3 access instead of long-lived access keys. Then `AWS_ACCESS_KEY_ID/AWS_SECRET_ACCESS_KEY` are not needed in the server `.env`.

## 5. CloudFront + OAC for a Private S3 Bucket

24. CloudFront -> Create distribution.
25. Choose the plan that fits the project (pricing/options can change over time).
26. Distribution name: bookkeepers-marketplace-cdn.
27. Origin type: Amazon S3.
28. S3 origin: bookkeepers-marketplace-prod-storage.
29. Keep Origin path blank so the whole bucket is the origin.
30. Enable private S3 bucket access from CloudFront.
31. Use Origin Access Control (OAC) and keep Sign requests enabled.
32. Allow AWS to update the bucket policy, or apply the generated policy.
33. Use HTTPS / redirect HTTP to HTTPS.
34. After deployment, copy the CloudFront domain.

```
AWS_PUBLIC_BASE_URL=https://d26wfj7v7cajxb.cloudfront.net
```

Object key: uploads/profile.jpg

Public URL: <https://d26wfj7v7cajxb.cloudfront.net/uploads/profile.jpg>

**Database tip:** Store the object key (for example uploads/profile.jpg) rather than a full S3 URL. This makes future CDN/domain changes much easier.

## 6. Launch the EC2 Server

| Setting | Recommended / Current                                                                                                |
|---------|----------------------------------------------------------------------------------------------------------------------|
| Region  | us-east-2 (Ohio) - same as S3                                                                                        |
| Name    | bookkeepers-marketplace-production                                                                                   |
| OS      | Current: Ubuntu 26.04 LTS x86_64; conservative future default: Ubuntu 24.04 LTS when dependencies are not yet tested |

| Setting  | Recommended / Current                  |
|----------|----------------------------------------|
| Instance | t3.small - 2 vCPU, 2 GB RAM            |
| Storage  | 30 GB gp3                              |
| Key pair | bookkeepers-marketplace-key, RSA, .pem |
| SSH      | TCP 22 -> My IP                        |
| HTTP     | TCP 80 -> Anywhere IPv4                |
| HTTPS    | TCP 443 -> Anywhere IPv4               |

35. EC2 -> Instances -> Launch instance.

36. Configure the instance using the settings above.

37. Do not publicly open 3000/3001/5000/27017/6379. Use Nginx and internal networking.

38. Launch the instance and wait for the instance state/status checks to pass.

**Resize later:** You can later move from t3.small to t3.medium/large: Stop instance -> Change instance type -> Start. EBS volumes can be increased, but reducing them is not straightforward.

## 7. Elastic IP - Stable Public IP

39. EC2 -> Network & Security -> Elastic IP addresses.

40. Allocate Elastic IP address.

41. Actions -> Associate Elastic IP address.

42. Resource type: Instance.

43. Instance: bookkeepers-marketplace-production.

44. Keep the automatically selected private IP and associate it.

Current project Elastic IP: 3.151.181.212

**Important:** Allocating an Elastic IP is not enough - it must be associated with the EC2 instance. If it is not associated, SSH can time out and DNS can point to the wrong address.

## 8. GoDaddy DNS Setup

| Type | Name  | Value         |
|------|-------|---------------|
| A    | @     | 3.151.181.212 |
| A    | admin | 3.151.181.212 |

| Type  | Name | Value                        |
|-------|------|------------------------------|
|       |      |                              |
| A     | api  | 3.151.181.212                |
| CNAME | www  | bookkeepersmarketplace.store |

- Do not change NS, SOA, \_domainconnect, or \_dmarc unless you have a specific reason.
- Root domain -> Frontend, admin subdomain -> Admin Dashboard, api subdomain -> Backend API.
- Verify with nslookup for the root, admin, and api hostnames.

## 9. SSH from Windows

**Ubuntu EC2 login:** No root password is provided. The default user is typically ubuntu and authentication uses the .pem private key.

```
ssh -i ".\bookkeepers-marketplace-key.pem" ubuntu@3.151.181.212
```

### Fix Windows .pem permissions - Command Prompt

```
icacls ".\bookkeepers-marketplace-key.pem" /inheritance:r
icacls ".\bookkeepers-marketplace-key.pem" /remove "NT AUTHORITY\Authenticated Users"
icacls ".\bookkeepers-marketplace-key.pem" /remove "BUILTIN\Users"
whoami
icacls ".\bookkeepers-marketplace-key.pem" /grant:r "%USERDOMAIN%\%USERNAME%:(R)"
icacls ".\bookkeepers-marketplace-key.pem"
```

### PowerShell variant

```
icacls ".\bookkeepers-marketplace-key.pem" /inheritance:r
icacls ".\bookkeepers-marketplace-key.pem" /remove "NT AUTHORITY\Authenticated Users"
icacls ".\bookkeepers-marketplace-key.pem" /remove "BUILTIN\Users"
icacls ".\bookkeepers-marketplace-key.pem" /grant:r "${env:USERDOMAIN}\${env:USERNAME}:(R)"
```

**If you change computer/country:** When SSH is restricted to My IP, update the Security Group inbound SSH source to your new current public IP before connecting.

## 10. Basic Ubuntu Server Setup

```
sudo apt update
sudo apt upgrade -y
sudo apt install -y git curl unzip nginx
sudo systemctl enable nginx
```

```
sudo systemctl start nginx
```

## Docker + Docker Compose

```
sudo apt install -y docker.io docker-compose-v2
sudo systemctl enable docker
sudo systemctl start docker
sudo usermod -aG docker ubuntu
exit
```

- Reconnect over SSH after logout so the docker group membership takes effect.
- Check: `docker --version`, `docker compose version`, `docker ps`.

## 2 GB Swap (useful for t3.small)

```
sudo fallocate -l 2G /swapfile
sudo chmod 600 /swapfile
sudo mkswap /swapfile
sudo swapon /swapfile
echo "/swapfile none swap sw 0 0" | sudo tee -a /etc/fstab
free -h
```

# 11. Frontend + Admin + Backend Deployment Structure

```
/home/ubuntu/bookkeepers-marketplace/
├── frontend/      # Next.js -> 3000
├── admin/         # Next.js -> host 3001
├── backend/       # NestJS -> 5000
└── docker-compose.yml
```

45. `mkdir -p ~/bookkeepers-marketplace && cd ~/bookkeepers-marketplace`
46. For private GitHub repositories, configure Deploy Keys or another secure GitHub authentication method.
47. Clone the Frontend, Admin, and Backend repositories.
48. Create separate `.env` / `.env.production` files.
49. Bind application host ports to 127.0.0.1 so they are not publicly exposed.

```
services:
  frontend:
    build: ./frontend
    restart: unless-stopped
    ports:
      - "127.0.0.1:3000:3000"

  admin:
    build: ./admin
    restart: unless-stopped
    ports:
      - "127.0.0.1:3001:3000"

  backend:
    build: ./backend
    restart: unless-stopped
    ports:
```

```
- "127.0.0.1:5000:5000"
```

## 12. Environment Variables - Safe Template

```
# Backend
NODE_ENV=production
PORT=5000
MONGODB_URI=...
JWT_SECRET=...

AWS_BUCKET_NAME=bookkeepers-marketplace-prod-storage
AWS_REGION=us-east-2
AWS_PUBLIC_BASE_URL=https://d26wfj7v7cajxb.cloudfront.net
# Only if your architecture still uses IAM access keys:
AWS_ACCESS_KEY_ID=...
AWS_SECRET_ACCESS_KEY=...

STRIPE_SECRET_KEY=sk_test_...
STRIPE_WEBHOOK_SECRET=whsec_...

FRONTEND_URL=https://bookkeepersmarketplace.store
ADMIN_URL=https://admin.bookkeepersmarketplace.store
```

```
# Frontend
NEXT_PUBLIC_API_URL=https://api.bookkeepersmarketplace.store
NEXT_PUBLIC_STRIPE_PUBLISHABLE_KEY=pk_test_...

# Admin
NEXT_PUBLIC_API_URL=https://api.bookkeepersmarketplace.store
NEXTAUTH_URL=https://admin.bookkeepersmarketplace.store
NEXTAUTH_SECRET=...
```

**Never commit:** .env, .env.production, \*.pem, AWS secret keys, Stripe secret keys, or database passwords.

## 13. Nginx Domain Routing

| Domain                             | Internal application |
|------------------------------------|----------------------|
| bookkeepersmarketplace.store + www | 127.0.0.1:3000       |
| admin.bookkeepersmarketplace.store | 127.0.0.1:3001       |
| api.bookkeepersmarketplace.store   | 127.0.0.1:5000       |

```
server {
    listen 80;
    server_name api.bookkeepersmarketplace.store;

    client_max_body_size 50M;
```

```

location / {
    proxy_pass http://127.0.0.1:5000;
    proxy_http_version 1.1;
    proxy_set_header Host $host;
    proxy_set_header X-Real-IP $remote_addr;
    proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
    proxy_set_header X-Forwarded-Proto $scheme;
}

```

- Use the same pattern for Frontend and Admin, changing only server\_name and proxy\_pass port.
- Enable the sites, run `nginx -t`, then reload Nginx.

```

sudo nginx -t
sudo systemctl reload nginx

```

## 14. SSL / HTTPS

```

sudo apt install -y certbot python3-certbot-nginx

sudo certbot --nginx -d bookkeepersmarketplace.store -d www.bookkeepersmarketplace.store
sudo certbot --nginx -d admin.bookkeepersmarketplace.store
sudo certbot --nginx -d api.bookkeepersmarketplace.store

sudo certbot renew --dry-run

```

**Before Certbot:** The DNS records must resolve to the EC2 Elastic IP and ports 80/443 must be open in the Security Group.

## 15. Connect MongoDB, S3, and Stripe

- **MongoDB Atlas:** Allow the EC2 Elastic IP /32 in Network Access. Avoid 0.0.0.0/0 where possible.
- **S3:** Keep the bucket private. Upload objects to S3 and generate read URLs from the CloudFront base URL.
- **Stripe:** Complete integration in test mode first. After the backend is live, configure a webhook endpoint such as `https://api.bookkeepersmarketplace.store/api/v1/stripe/webhook`.
- **Stripe live mode:** After the client completes business verification, configure `pk_live/sk_live` and the separate live webhook signing secret.

## 16. Common Problems and Quick Fixes

| Problem                      | Likely reason / fix                                                                                                           |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| SSH timeout                  | Elastic IP is not associated, port 22 is blocked, or the My IP source is stale.                                               |
| UNPROTECTED PRIVATE KEY FILE | The Windows ACL for the .pem file is too open. Use <code>icacls</code> to remove inheritance/group access and grant only your |

| Problem                           | Likely reason / fix                                                                                                                                           |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                   | user Read access.                                                                                                                                             |
| Permission denied (publickey)     | Wrong key, wrong username, or key-file permission issue. Ubuntu AMIs typically use ubuntu.                                                                    |
| S3 direct URL AccessDenied        | Expected for a private bucket. Use the CloudFront OAC URL.                                                                                                    |
| CloudFront NXDOMAIN               | Distribution is still deploying or the hostname is mistyped. Wait for deployment, copy the exact domain, and optionally run <code>ipconfig /flushdns</code> . |
| Domain resolves but app fails     | Check Nginx config, container status, internal ports, <code>.env</code> values, and application logs.                                                         |
| SSH fails from another PC/country | Update the Security Group SSH source to the new My IP.                                                                                                        |

## 17. Final Production Checklist

- Root MFA enabled; developer does not have root credentials.
- Separate developer IAM access; permissions reduced after setup.
- S3 private; Block Public Access enabled; CloudFront OAC configured.
- EC2 Security Group: 22 from My IP, 80/443 public; app/database/Redis ports not public.
- Elastic IP associated and GoDaddy DNS correct.
- Docker containers have restart policies.
- Nginx routing works for root/admin/api domains.
- HTTPS certificates installed and renewal test passes.
- MongoDB network restrictions configured.
- Secrets are not in Git; the `.pem` key is stored securely.
- AWS Budget/Cost alerts configured.
- Unused Elastic IPs, EC2 instances, and storage resources removed.

**Cost reminder:** EC2, EBS storage, public IPv4/Elastic IP, S3 storage/requests, and CloudFront usage can generate charges. Remove unused resources and configure a budget alert.